

Encoding Strategies for Multimodal Medical VQA and Continuous Robot Control

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Abstract—This report investigates encoding strategies for two machine learning tasks: (1) Visual Question Answering (VQA) on the SLAKE medical dataset, and (2) robot locomotion control on Walker2d-v5 using deep reinforcement learning. For Task 1, over ten multimodal classifiers were trained, varying vision encoders (ViT, ResNet50, EfficientNet-B4, BiomedCLIP), text encoders (DistilBERT, BiomedBERT), and fusion strategies (concatenation, Hadamard product, cross-attention, concat+similarity). An agentic AI framework using LangGraph searched 24 hyperparameter configurations. BiomedCLIP achieved the best accuracy (80.77%) and balanced accuracy (57.50%), demonstrating that biomedical pretraining outperforms general-purpose encoders. For Task 2, PPO, SAC, and TD3 were compared using MLP and 1D CNN state representations across multiple seeds. SAC with MLP achieved the highest reward (5142.98), while TD3 with CNN reached 4078 on its best seed but required double the training budget and showed higher variance overall.

I. TASK 1: VISUAL QUESTION ANSWERING

A. Problem Description

Given a medical image and a natural language question, the task is to predict the correct answer from all unique answers in the training data (228 classes). The SLAKE dataset [1] contains 642 medical images (CT, MRI, X-ray) with ~4,900 English training pairs and 1,061 test pairs. The dataset exhibits significant class imbalance: “Yes” and “No” account for 34% of training samples, while 60 answer classes appear only once. Answers unseen during training receive label -1 and are always counted as errors.

B. Methodology

All models follow a late-fusion architecture adapted from workshop materials [2]: a vision encoder produces image features, a text encoder produces question features, and a fusion module combines them before classification. Three axes of variation were explored.

Vision encoders: ViT-B/32 [5] (Transformer, 88M params), ResNet50 [8] (CNN, 25M params), EfficientNet-B4 [7] (CNN, 19M params, 380×380 input), and BiomedCLIP [6] (ViT pretrained on 15M biomedical image-text pairs).

Text encoders: DistilBERT (66M params, general English) and BiomedBERT (110M params, PubMed abstracts).

Fusion strategies: Concatenation, Hadamard product (element-wise multiplication), cross-attention (image queries attend to question tokens), and concat+similarity (concatenation with cosine similarity score). A CLIP contrastive loss variant was also tested.

All models were trained with AdamW (weight decay = 10^{-4}), CrossEntropyLoss, early stopping (patience = 5), and selective layer unfreezing (2–4 vision blocks) to balance adaptation against overfitting.

C. Results

Table I presents results on the held-out test set. Models marked (A) were discovered by the agentic AI framework.

TABLE I: VQA results on SLAKE. All values in %. Best in bold.

Model	Acc	Bal.Acc	F1	MRR	ECE	Time(s)
BiomedCLIP + C+S	80.77	57.50	80.72	88.12	0.038	2934
ResNet + Had. (A)	80.02	54.02	78.99	87.39	0.072	1605
ViT + Hadamard	79.83	54.79	79.07	87.24	0.059	1661
EffNet + Hadamard	79.55	52.31	79.22	87.23	0.079	2942
ViT + BiomedBERT + Had.	79.08	53.83	78.72	86.85	0.086	1791
ViT + Concat	78.98	52.85	78.52	86.74	0.059	1483
ResNet + Concat (A)	78.89	55.66	78.70	86.69	0.062	892
ViT + CrossAttn	75.12	46.65	73.43	83.79	0.064	1711
ViT + CLIP Loss	63.52	17.57	57.45	73.08	0.100	2267
BiomedCLIP + Had.	36.00	4.36	25.52	48.81	0.134	2862

D. Analysis

Fusion strategy. Hadamard fusion outperformed concatenation in every encoder pairing (e.g., ViT+Hadamard 79.83% vs ViT+Concat 78.98%). The element-wise product acts as mutual gating: a feature must be activated in both modalities to survive, suppressing irrelevant dimensions. BiomedCLIP’s concat+similarity fusion achieved the best results by adding an explicit cosine alignment signal alongside the raw features.

Vision encoder. ViT outperformed both CNNs by 1–3%, indicating that global patch-level attention captures diagnostically relevant context (scan modality, organ location) more effectively than local convolutions. BiomedCLIP’s balanced accuracy (57.50%) exceeded all other models, confirming that encoders pretrained on biomedical data produce features that ImageNet pretraining cannot replicate. The gap between accuracy (~80%) and balanced accuracy (~55%) across all models reflects the class imbalance: models predict frequent answers (“Yes”, “No”, “Lung”) well but struggle with the 60 classes that appear only once in training.

Text encoder. BiomedBERT with Hadamard fusion did not improve over DistilBERT with the same fusion (79.08% vs 79.83%). However, this comparison is not fully controlled as both runs used different random initialisations. The limited

improvement likely reflects that SLAKE questions are linguistically simple (e.g., “What modality is used?”), reducing the benefit of medical vocabulary.

Failed approaches. CLIP contrastive loss scored 63.52% because SLAKE reuses images across questions, producing contradictory gradients in the $N \times N$ similarity matrix. Biomed-CLIP with Hadamard scored only 36% (see Table I) because its L2-normalised features yield near-zero products when multiplied element-wise, collapsing the information signal.

E. Agentic AI Framework

A LangGraph-based agentic framework was adapted from the workshop, following the brief’s recommendation for systematic experimentation under the agentic AI paradigm. The state machine cycles: perceive $\rightarrow$ select action $\rightarrow$ run experiment $\rightarrow$ update history, replacing the workshop’s simulated experiments with actual model training and evaluation.

The framework searched over 6 model types, 2 learning rates, and 2 unfreezing depths (24 configurations total). While this grid is not exhaustive, it enabled controlled comparison across all combinations something impractical to do manually.

The results confirmed that 3×10^{-5} (a standard fine-tuning rate) consistently outperformed 1×10^{-5} across all architectures, and identified ResNet+Hadamard with 4 unfrozen stages (80.02%) as a competitive configuration that had not been tested in manual experiments. BiomedCLIP remained the overall best model, which the agentic search did not include due to its separate pipeline.

II. TASK 2: ROBOT POLICY LEARNING

A. Problem Description

Walker2d-v5 [4] is a MuJoCo environment where a bipedal robot learns forward locomotion. Observations are 17-dimensional vectors (joint angles, velocities, torso orientation); actions are 6-dimensional continuous torques. The reward sums forward velocity, a healthy bonus (+1 per step while upright), and a control cost. Episodes last up to 1,000 steps.

B. Methodology

Three algorithms were compared: **PPO** [11] (on-policy, clipped surrogate objective), **SAC** [9] (off-policy, entropy-regularised, twin critics, auto-tuned temperature), and **TD3** [10] (off-policy, twin critics, delayed policy updates). All used `MlpPolicy` from StableBaselines3 [3], trained for 1.5M timesteps with 3 seeds each (77, 200, 431).

A **1D CNN** feature extractor was implemented using `Conv1d` layers to detect local patterns across the observation vector. CNN agents trained with separate seeds (55, 234, 678) for 3M timesteps; different seeds from MLP were used to avoid bias, though this introduces additional variance in the comparison. A **Transformer** variant using self-attention was also designed; initial SAC runs revealed instability (critic loss 7.57×10^{24}) due to unnormalised inputs. This was resolved by adding LayerNorm at input and output, but full multi-seed results were not obtained due to time constraints.

C. Results

Table II shows averaged results. Fig. 1 shows the training learning curves.

TABLE II: DRL agent comparison on Walker2d-v5. Best in bold.

Algorithm	Arch	Avg Reward	Avg Steps	Timesteps
SAC	MLP	5142.98	987.6	1.5M
TD3	MLP	4423.32	956.9	1.5M
SAC	CNN	3343.07*	806.4	3M
TD3	CNN	2890.34	751.8	3M
PPO	MLP	2068.64	586.5	1.5M
PPO	CNN	596.92	278.8	600K

*Single seed; remaining seeds did not complete training within available compute time.

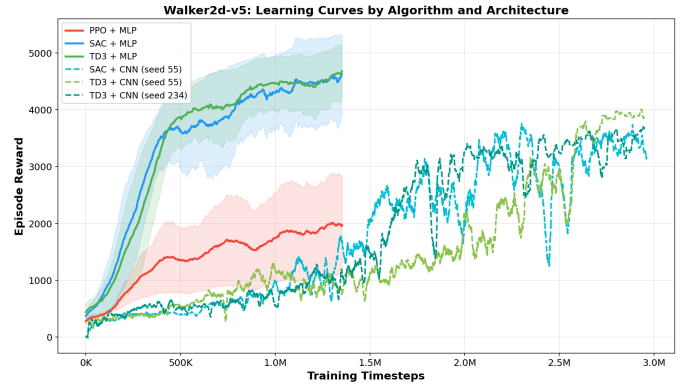


Fig. 1: Walker2d-v5 learning curves. Solid lines: MLP agents averaged across 3 seeds (shaded = ± 1 std). Dashed lines: individual CNN seeds.

D. Analysis

Algorithm comparison. SAC achieved the highest reward (5142.98), with two seeds sustaining the maximum 1,000 steps throughout evaluation - the robot walked entire episodes without falling. Its entropy regularisation drives exploration of diverse gaits before convergence. TD3 ranked second (4423.32), benefiting from twin critics that prevent Q-value overestimation. PPO ranked last (2068.64): as an on-policy algorithm it discards experience after each update, making it less sample-efficient. One PPO seed scored only 1,098, highlighting initialisation sensitivity.

State representation. CNN produced mixed results. TD3+CNN seed 55 achieved 4078 reward with 1,000 average steps, comparable to TD3+MLP. However, TD3+CNN seed 678 scored only 1,067, and CNN agents required 3M timesteps. Walker2d’s observation vector has no spatial ordering, a joint’s position and velocity can be many indices apart, so CNN’s local-neighbour assumption adds learning difficulty without useful structure. MLP imposes no ordering and proved more efficient and reliable.

Recommendation. SAC with MLP is recommended for Walker2d-v5. Off-policy replay enables sample-efficient learning, and entropy regularisation prevents premature convergence to suboptimal gaits.

III. CONCLUSION

Across both tasks, encoding strategy proved more consequential than architectural complexity. In medical VQA, BiomedCLIP’s biomedical pretraining delivered the best balanced accuracy, outperforming all manually engineered and agentic-discovered configurations. The agentic framework nonetheless identified learning rate as the dominant hyperparameter and found competitive configurations absent from manual search.

In robot control, SAC with MLP outperformed all alternatives including CNN agents given double the training budget. CNN could reach competitive performance on individual seeds but at greater computational cost and lower reliability. Both tasks reinforce that inductive biases must match data structure; more complex architectures are not inherently better.

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